GradLink Authentication Architecture Report

Project: GradLink (Django REST Framework)

Date: March 3, 2026

Scope: Authentication mechanism analysis, comparison, and justification

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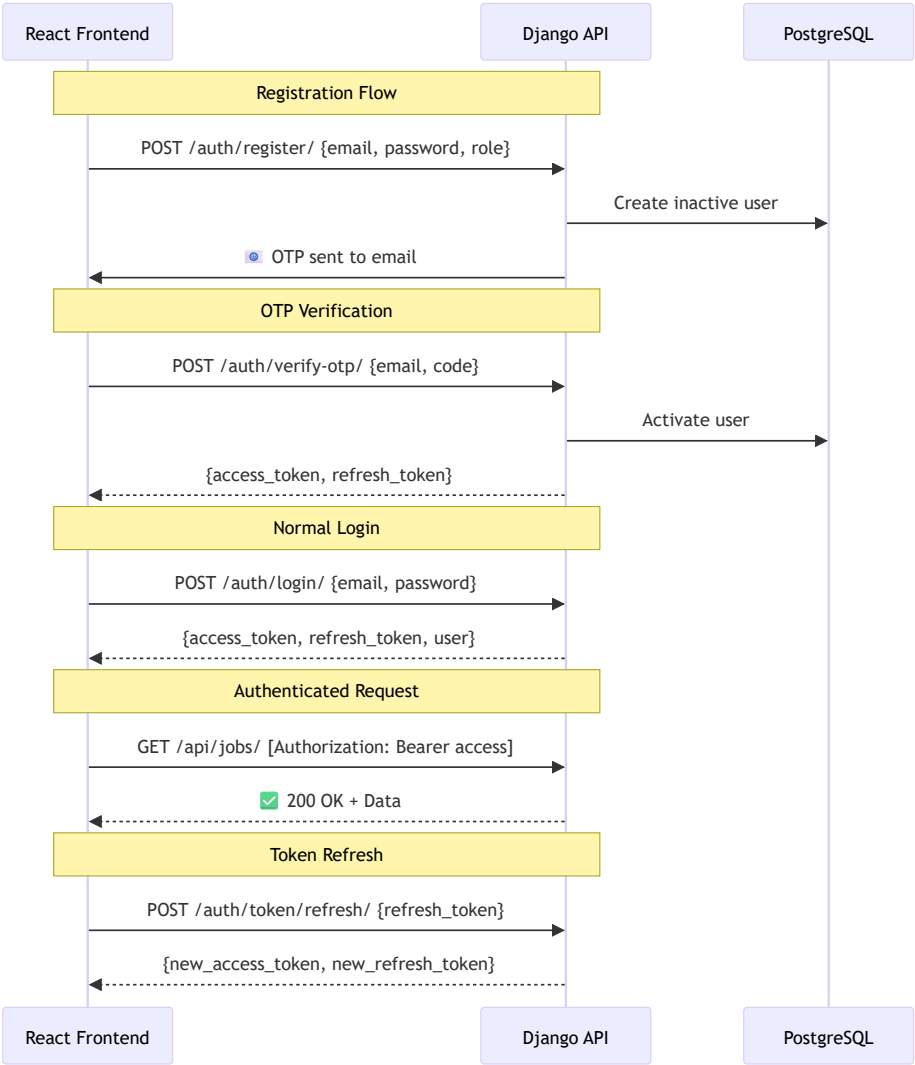
1. Authentication Type Used: JWT (JSON Web Tokens) via SimpleJWT

GradLink uses `djangorestframework-simplejwt` as its primary authentication backend. This is a **stateless, token-based authentication system** that issues two types of tokens:

Token	Purpose	Lifetime
Access Token	Sent with every API request in the `Authorization: Bearer <token>` header to prove identity	**60 minutes**
Refresh Token	Used to obtain a new access token without re-entering credentials	**1 day**

How It Works in GradLink



Configuration in settings.py

```
REST_FRAMEWORK = {
    'DEFAULT_AUTHENTICATION_CLASSES': (
        'rest_framework_simplejwt.authentication.JWTAuthenticat
    ),
}

SIMPLE_JWT = {
    'ACCESS_TOKEN_LIFETIME': timedelta(minutes=60),
    'REFRESH_TOKEN_LIFETIME': timedelta(days=1),
    'ROTATE_REFRESH_TOKENS': True,        # New refresh token o
    'BLACKLIST_AFTER_ROTATION': True,      # Old refresh tokens
}
```

Additional Auth Features Implemented

Feature	Implementation
Email-based login	Custom User model with `email` as `USERNAME_FIELD` (no username)
OTP Email Verification	6-digit OTP sent via SMTP, stored in `OTPVerification` model with expiry & attempt limits
Google OAuth2 SSO	Server-side Google token verification → auto-create/login user → issue JWT
Role-based access	User roles: `student`, `employer`, `admin`
Soft delete	`soft_delete()` / `reactivate()` methods with 30-day window
Token rotation	Refresh tokens are rotated and old ones blacklisted

2. Comparison with Other Authentication Methods

2.1 Session-Based Authentication (Django Default)

Aspect	Session Auth	JWT (SimpleJWT) ✅
State	**Stateful** — session stored server-side (DB / cache)	**Stateless** — token is self-contained
Storage	Server stores session in DB; client holds `sessionid` cookie	Server stores nothing*; client holds token
Scalability	❌ Every request hits DB to validate session	✅ Token validated by signature only — no DB hit
Cross-domain	❌ Cookies are domain-locked; hard with separate frontend	✅ Token sent in header; works across any domain
Mobile / SPA	❌ Cookies awkward on mobile & single-page apps	✅ Perfect fit for React SPA + mobile apps
CSRF	❌ Requires CSRF protection for cookies	✅ No cookies = no CSRF vulnerability
Logout	✅ Simple server-side invalidation	⚠️ Requires blacklisting (GradLink has this)

> * With `BLACKLIST_AFTER_ROTATION=True`, GradLink does store blacklisted refresh tokens, but this is a lightweight operation.

2.2 Token Authentication (DRF's Built-in TokenAuthentication)

Aspect	DRF TokenAuth	JWT (SimpleJWT) ✅
Token type	Single permanent token per user	Paired access + refresh tokens
Expiration	❌ **No built-in expiration** — token is valid forever	✅ Access token auto-expires (60 min)
DB dependency	❌ Every request queries the `authtoken_token` table	✅ Access token verified by cryptographic signature
Security	❌ If stolen, token is valid indefinitely	✅ Short-lived access token limits damage window
Rotation	❌ Manual implementation needed	✅ Built-in refresh rotation + blacklisting
Payload	❌ No payload — just an opaque string	✅ JWT contains claims (user ID, role, expiry)

2.3 OAuth2 / Third-Party Providers Only

Aspect	OAuth2 Only	JWT + OAuth2 (GradLink's approach) ✅
Provider dependency	❌ 100% dependent on Google/Facebook uptime	✅ JWT works independently; Google is optional
Local accounts	❌ Cannot support email/password registration	✅ Supports both local + Google sign-in
Complexity	❌ Complex OAuth flows for every request	✅ OAuth only at login; JWT for all subsequent requests
Offline access	❌ Requires constant provider connectivity	✅ JWT validated locally

2.4 API Key Authentication

Aspect	API Key	JWT (SimpleJWT) ✅
Use case	Machine-to-machine / 3rd party integrations	User-facing applications
Identity	❌ Identifies the application, not the user	✅ Token carries user identity (ID, role)
Expiration	❌ Typically permanent	✅ Auto-expiring with rotation
Granularity	❌ One key per service	✅ Per-user tokens with role-based permissions

2.5 Summary Comparison Matrix

Feature	Session	DRF Token	API Key	OAuth2 Only	JWT (SimpleJWT) ✅
Stateless	❌	❌	❌	❌	✅
Auto-expiration	✅	❌	❌	✅	✅
SPA-friendly	❌	✅	✅	⚠️	✅
Mobile-friendly	❌	✅	✅	⚠️	✅
No CSRF needed	❌	✅	✅	✅	✅
Carries user data	❌	❌	❌	✅	✅
Scalable	❌	❌	✅	⚠️	✅
Works cross-domain	❌	✅	✅	✅	✅

3. Why JWT (SimpleJWT) Is the Best Fit for GradLink

3.1 Architecture Match: Django + React SPA

GradLink's architecture is a **Django REST API backend** with a **React frontend** running on separate domains/ports (`localhost:8000` vs `localhost:3000`). This is the textbook use case for JWT:

- **Cross-origin by design** — JWT tokens are sent via `Authorization` headers, not cookies, so CORS is straightforward.
- **No session infrastructure needed** — The React app stores tokens in memory/localStorage and attaches them to every request.
- **No CSRF vulnerabilities** — Since no cookies are used for auth, CSRF attacks are not applicable.

3.2 Performance & Scalability

- **Zero DB queries for authentication** — Every incoming request is validated by verifying the JWT signature using the `SECRET_KEY`. No database roundtrip.
- **Horizontal scaling** — Any server instance can validate any token. No shared session store (e.g., Redis) needed for auth.
- **Ideal for high-traffic endpoints** — Job listings, ATS operations, and profile queries are authenticated without DB overhead.

3.3 Security Features Already Implemented

GradLink has leveraged SimpleJWT's security features correctly:

Security Feature	Status	Impact
Short access token lifetime (60 min)	✅ Configured	Limits exposure if token is stolen
Refresh token rotation	✅ `ROTATE_REFRESH_TOKENS=True`	Every refresh issues a new pair, preventing reuse
Blacklisting old tokens	✅ `BLACKLIST_AFTER_ROTATION=True`	Stolen refresh tokens become invalid after rotation
OTP email verification	✅ Custom implementation	Prevents fake accounts; verifies email ownership
Google OAuth2 integration	✅ Server-side verification	Trusted identity from Google without storing Google passwords
Password validation	✅ Django validators enabled	Enforces strong passwords

3.4 Multi-Role System Compatibility

GradLink has three distinct user roles: `student`, `employer`, and `admin`. JWT naturally supports this because:

- The token payload carries the user ID, which maps to a role in the database.

- Role-based permission classes can be applied per-view without additional auth queries.
- Different frontends (student dashboard, employer portal, admin panel) can all use the same JWT mechanism.

3.5 Mobile-Ready

If GradLink ever expands to mobile apps (iOS/Android), JWT works out of the box:

- Tokens are stored in secure device storage.
- No cookie management or WebView session hacks.
- Same API, same auth headers, same backend — zero changes needed.

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4. Potential Improvements

> 💡 **TIP**
> These are recommendations for further hardening, not issues with the current choice.

Improvement	Rationale
Store tokens in `httpOnly` cookies (optional)	Prevents XSS from accessing tokens in localStorage. SimpleJWT supports this pattern.
Add `jti` claim checking	Track individual token IDs for per-token revocation.
Implement sliding sessions	Automatically extend active users' sessions by issuing new tokens on activity.
Add rate limiting on login/OTP	Prevent brute-force attacks on the login and OTP endpoints.
Consider `RS256` signing	Allows public key verification by microservices without sharing the secret key.

5. Conclusion

JWT via SimpleJWT is the optimal authentication strategy for GradLink because:

1. ✅ It is **purpose-built** for Django REST + React SPA architectures.
2. ✅ It provides **stateless, scalable authentication** with zero DB overhead per request.
3. ✅ It supports **cross-origin requests** natively — essential for separate frontend/backend deployments.
4. ✅ The **token rotation and blacklisting** features are already configured, providing strong security.
5. ✅ It seamlessly integrates with **Google OAuth2 SSO** as an additional login method.
6. ✅ It's **mobile-ready** for future platform expansion.
7. ✅ It's the **industry standard** for modern API authentication, with strong community support and documentation.

No other authentication method checks all these boxes for GradLink's specific requirements.